

I. Sensorless Situation

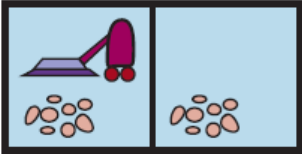
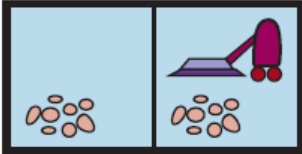
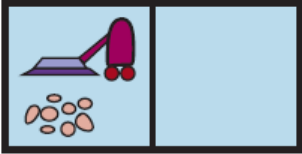
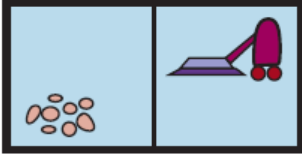
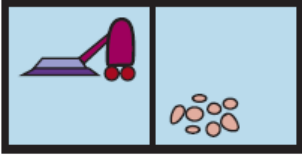
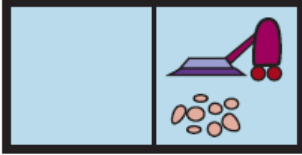
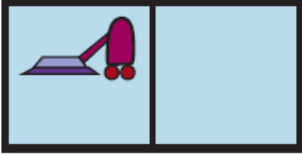
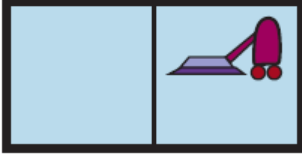
Agent's precepts provide no information at all.

- ♦ Sensorless solutions are surprisingly common.
- ♦ They can be *robust* due to zero dependence on sensors working properly.

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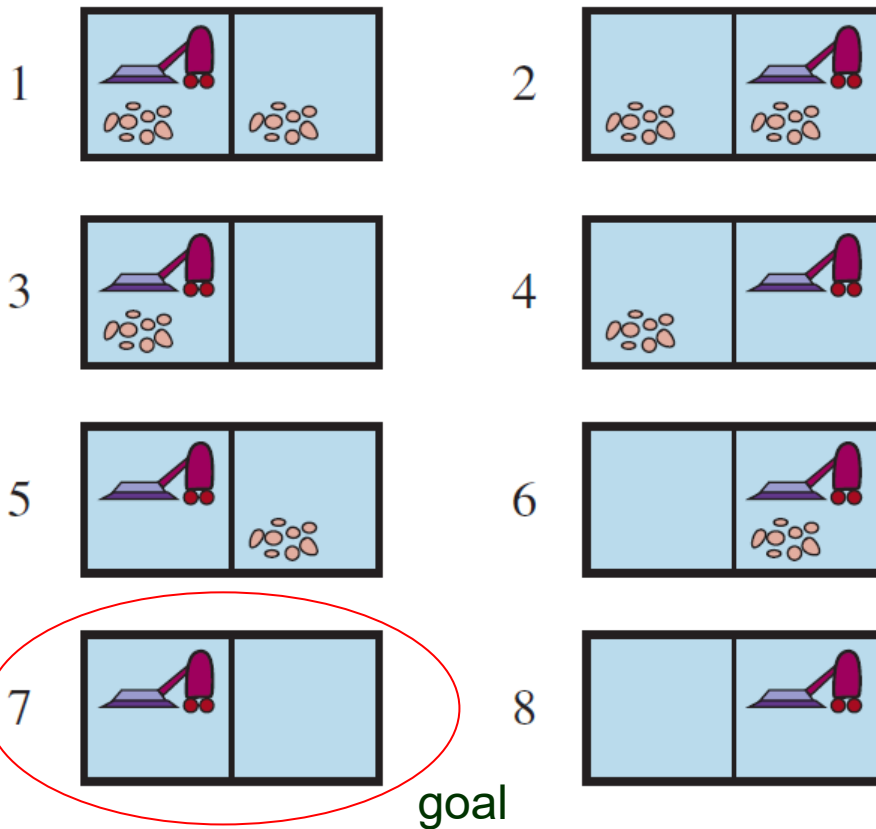
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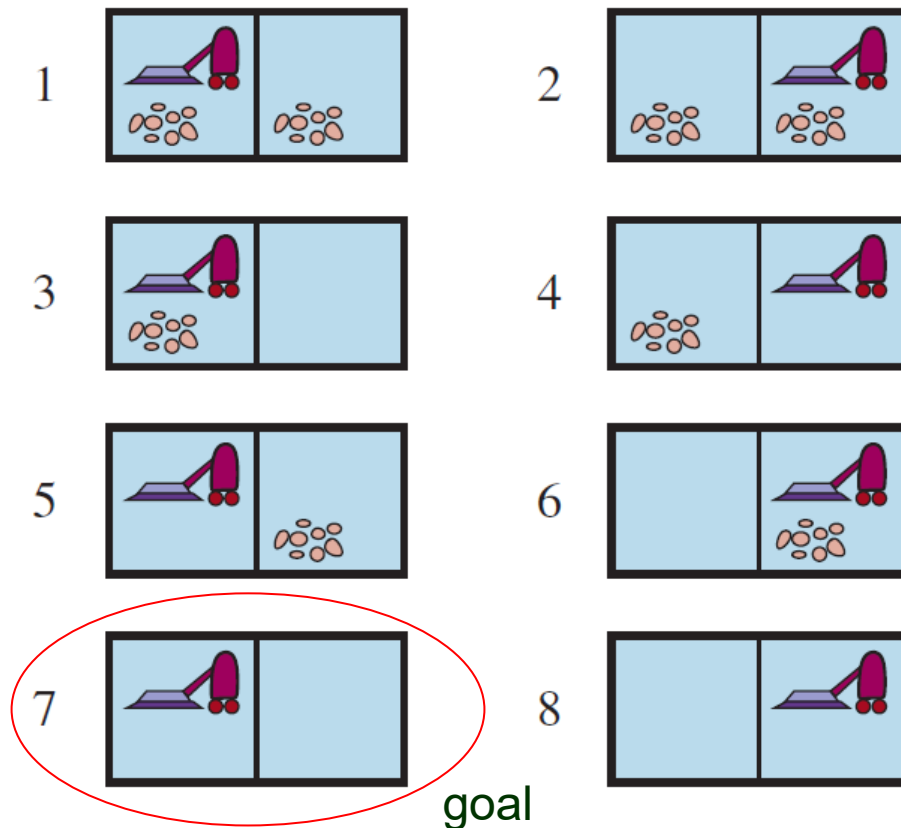
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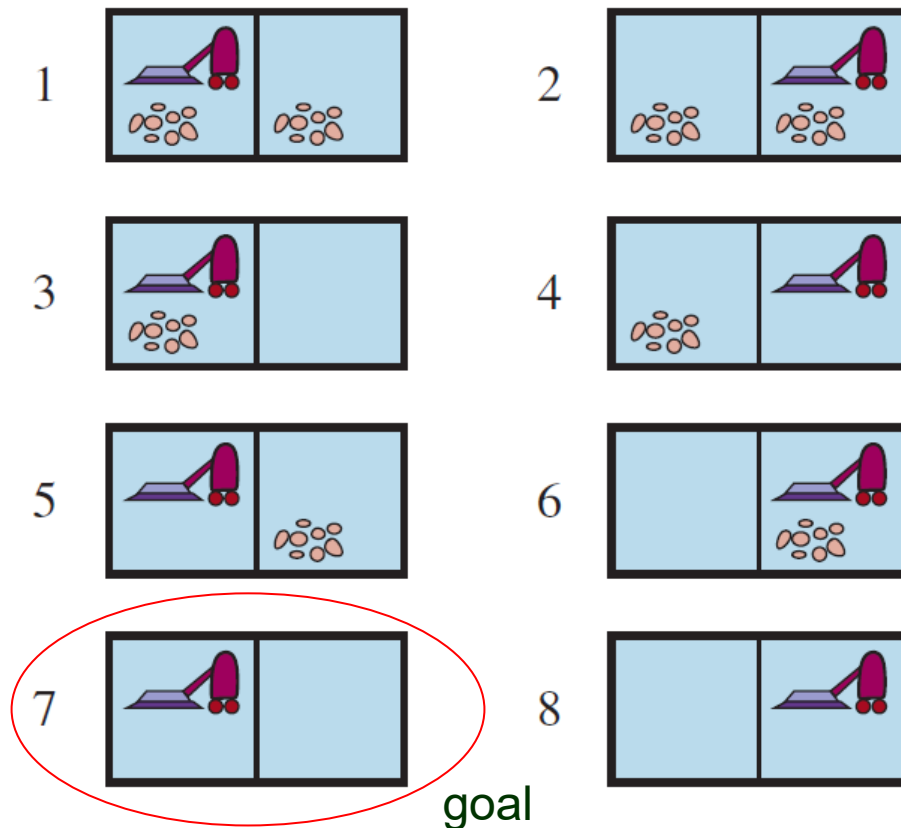
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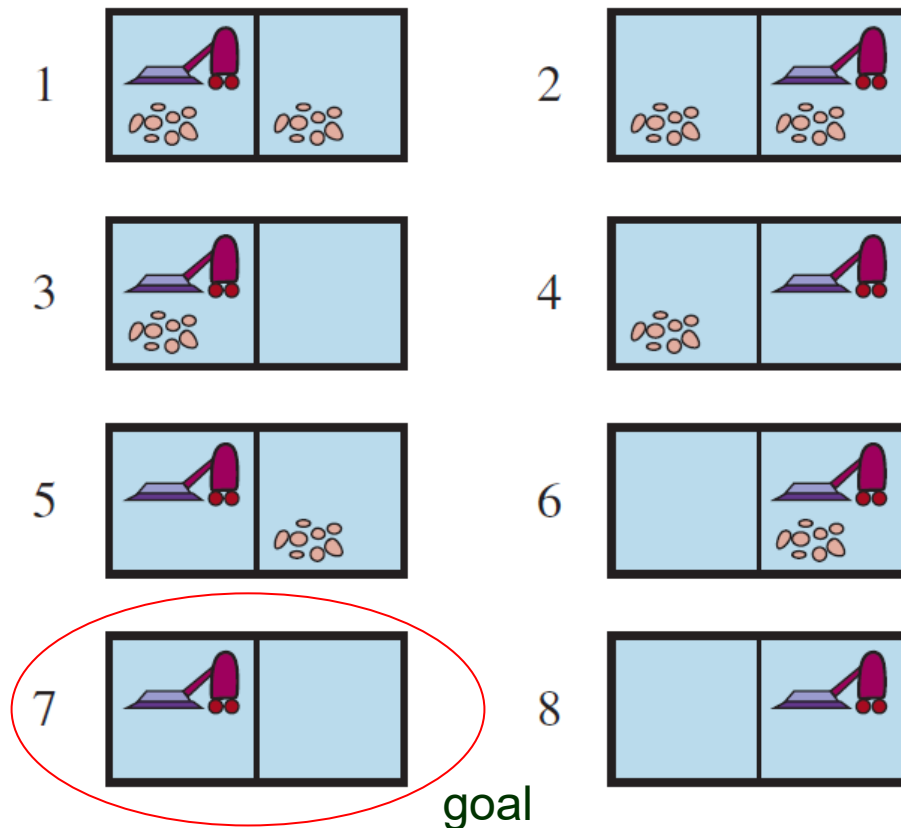
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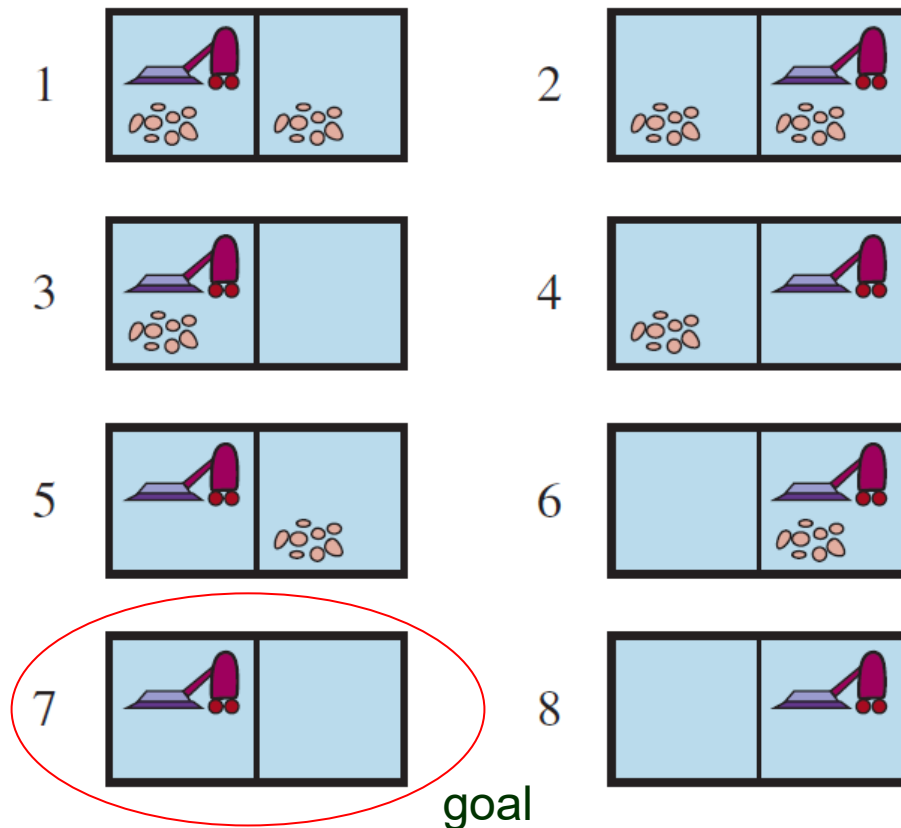
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Solution to a Sensorless Problem

A sequence of actions from search in the space B of *belief states* (*b-states*).

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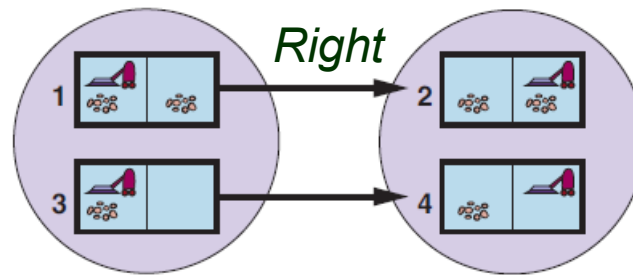
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↑
Action on a physical state

Transition Model

- **Transition model:** The new belief state is, under deterministic actions,

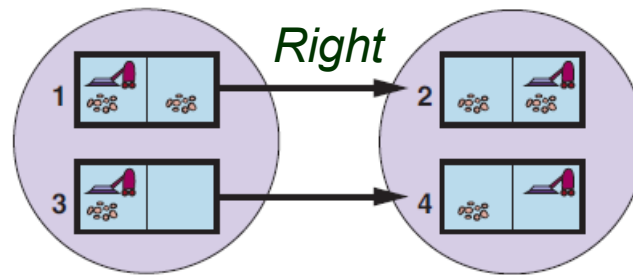
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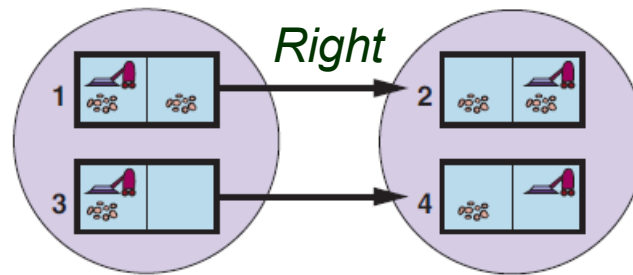


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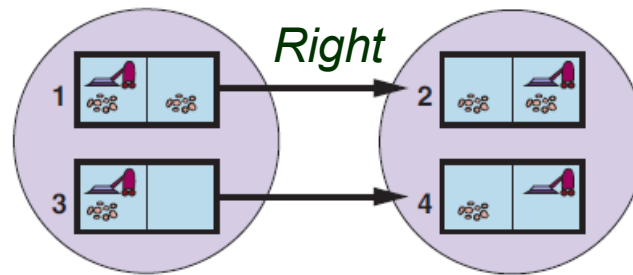
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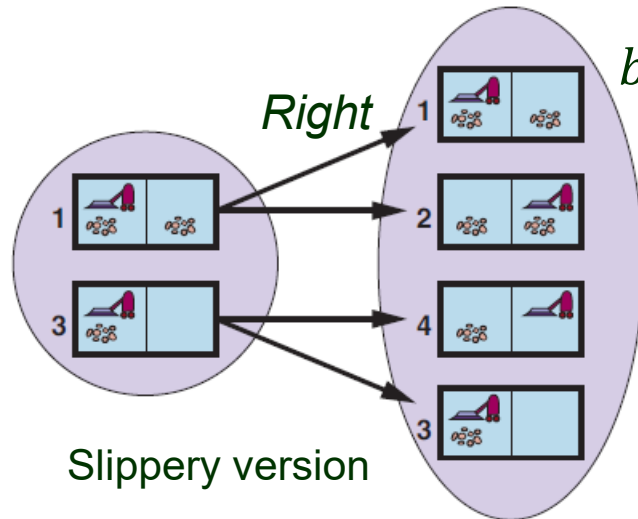
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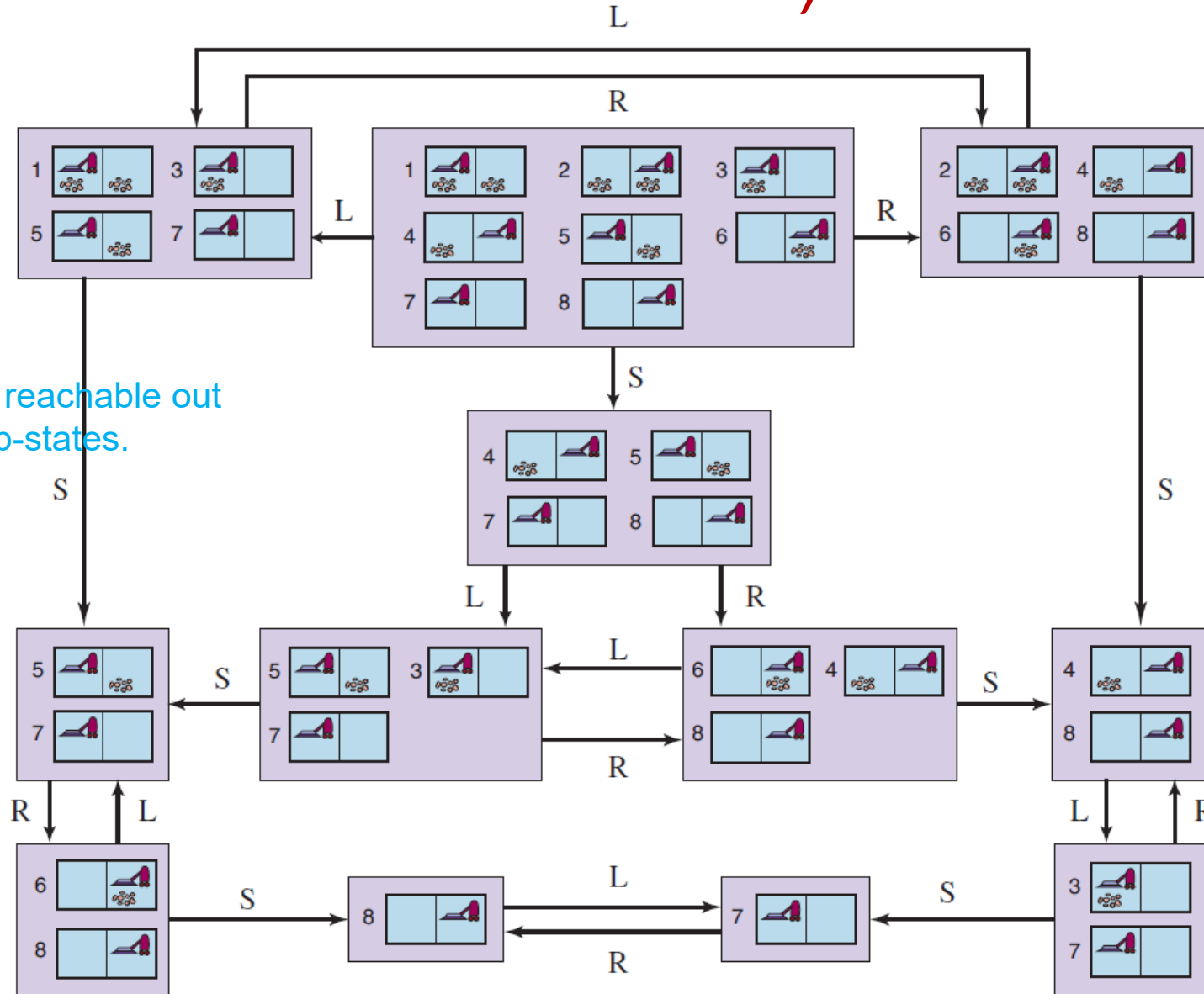
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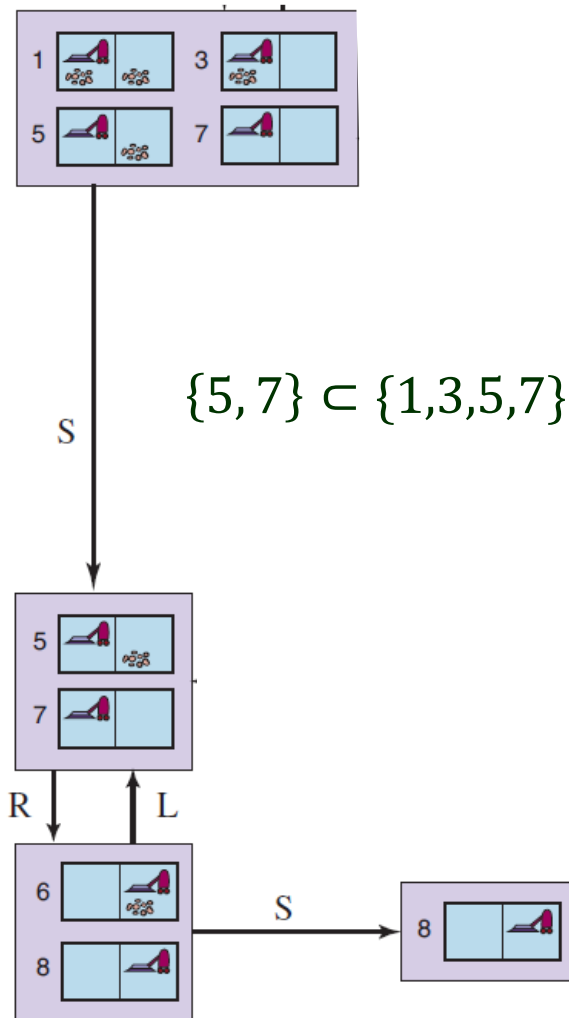
Goal & Cost of Action

- Goal test: The goal is achieved
 - ♣ *possibly* if one of the states $s \in b$ passes the test;
 - ♣ *necessarily* if every state $s \in b$ passes the test.
- Action cost: Could be one of several values if the same action has different costs in different states.

Reachable Belief States (Sensorless & Deterministic)



Subset and Superset States



♦ Prune a **superset** b-state, say, $\{1, 3, 5, 7\}$, to concentrate on solving the easier subset b-state, say, $\{5, 7\}$.

♦ Prune a **subset** b-state, say, $\{2, 5, 7\}$, if a superset b-state, say, $\{1, 3, 5, 7\}$, of the resulting subset, say, $\{5, 7\}$, has been found to be solvable already.

Handling Search Space Complexity

N physical states $\Rightarrow 2^N$ belief states!

- Use of compact description, e.g., mathematical logic (Chapter 7)
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- ♣ Check if it works for state 2.

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- If yes, check if it works for state 3, and so on.

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- ♣ Find an action sequence that works for state 1.

- ♣ Check if it works for state 2.

- If yes, check if it works for state 3, and so on.
 - If no, find a different solution for state 1, and so on.